

# Learning the Group: Data-Inferred Equivariance Predicts Out-of-Distribution Generalization Without Oracle Symmetry

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## Abstract

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A central limitation of recent symmetry-and-generalization work — including our own prior result that *weakness* (symmetry-compatible-hypothesis volume) predicts OOD generalization on partially-observed cyclic symmetries — is that the transformation group is supplied by an oracle. Perin and Deny [4] prove that conventional networks cannot extrapolate partially-observed cyclic symmetries; the natural follow-up question is whether the group can be inferred from training data alone. We give a positive answer for the finite-group case. Our `infer_rotation_group_from_training` procedure searches a discrete candidate set of transformations and keeps those under which training inputs map to other training inputs of the same label with high feature similarity. On a partially-observed  $Z_8$  rotation task with 256 trained models, the procedure recovers the true cyclic group with mean recall **89.7%** and precision **71.3%** from training data alone. Then, scoring learned-function weakness against this *data-inferred* group rather than the oracle, we find  $\text{Pearson } r(\text{weakness\_learned}, \text{OOD}) = \mathbf{+0.662}$  vs  $r(\text{weakness\_oracle}, \text{OOD}) = \mathbf{+0.736}$  — the data-inferred selector retains 90% of the oracle's signal. Both dominate every classical predictor: parameter  $L_2$  (+0.43), training accuracy (+0.42), training loss (−0.39), Hutchinson sharpness (+0.22). We report an honest negative finding for one control: a random-rotation null under a dense candidate set inherits  $Z_8$  structure passively and is not a strict null ( $r = +0.55$ ). The pixel-permutation control from our prior paper remains the clean null ( $r = -0.34$ ). We release the benchmark, transformation-generator code, Modal entrypoint, and 4 unit tests pinning recovery, identity preservation, and invariance scoring.

## 1. Introduction

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The weakness-invariance program [2] demonstrates that *symmetry-compatible-hypothesis volume* predicts OOD generalization where training loss, MDL, simplicity, flatness, and held-out validation cannot. The result is robust but has one critical limitation: the candidate transformation group is supplied by an oracle. In any real ML setting, the symmetry is exactly the thing the system needs to discover; assuming oracle access defeats the purpose.

This paper closes that gap for the finite-group case. We give a procedure that **infers the symmetry group from training data alone** and show that the resulting selector predicts OOD generalization almost as well as the oracle on a partial-orbit rotation task. The procedure is intentionally minimal: it searches a discrete set of candidate transformations and accepts those under which the training set is approximately self-consistent (label-preserving). No oracle group, no architectural equivariance, no learned-equivariance module from prior work [6]. The result is that the data alone — in the partial-orbit regime that defeats conventional supervised networks per Perin and Deny [4] — already contains enough signal to recover the cyclic group at the granularity needed for the weakness selector.

Contributions:

1. A simple, reproducible transformation-discovery procedure (`infer_rotation_group_from_training`) that recovers the true rotation group with **89.7% recall and 71.3% precision** from training data alone across 256 trained models.

2. A 256-model Modal-parallel neural sweep showing that learned-group weakness (no oracle) gives Pearson  $r = +0.662$  with OOD vs oracle  $r = +0.736$  — **90% of the oracle's predictive signal**, both dominating every classical predictor.
3. An honest finding: a random-rotation control under a *dense* candidate set is not a strict null ( $r = +0.55$ ) because random rotations from a 24-angle grid accidentally cover much of  $Z_8$  at our matching tolerance. The pixel-permutation control from [2] ( $r = -0.34$ ) remains the clean null.

## 2. Method

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### 2.1 Setup

We reuse the partial-orbit rotated-stroke benchmark from [2]: 8 stroke-pattern classes on  $16 \times 16$  grayscale images,  $Z_8$  cyclic rotation group, each class shown at only 3 of the 8 rotations during training. The held-out 5 rotations per class form the OOD set.

### 2.2 Transformation discovery

Given training inputs  $(x_i, y_i)$  and a discrete candidate angle set  $\Theta = \{0^\circ, \frac{360^\circ}{K}, \frac{2 \cdot 360^\circ}{K}, \dots, \frac{(K-1) \cdot 360^\circ}{K}\}$  for grid size  $K$ , we score each candidate angle  $\theta$  by

$$s(\theta) = \frac{1}{N} \sum_{i=1}^N \max_j |y_j - y_i| \cdot \text{flatten}(R_\theta(x_i)) \cdot \text{flatten}(x_j)$$

where  $R_\theta$  is image rotation by  $\theta$  and "flatten" is the raw pixel-vector. We keep angles with  $s(\theta) \geq \tau$  for a chosen threshold  $\tau \in [0, 1]$  and force the identity  $\theta = 0^\circ$  into the kept set. The kept set is the *learned group*.

The intuition: a rotation that respects the dataset's intrinsic symmetry will map every training input to *some* other training input of the same class, because both lie on the same orbit. A rotation that does not respect the symmetry will map training inputs to nowhere in the training set. Pixel-space cosine similarity adjudicates.

With  $K = 24$  candidate angles ( $15^\circ$  resolution) and  $\tau = 0.5$ , the procedure consistently recovers all 8 oracle  $Z_8$  angles  $\{0^\circ, 45^\circ, \dots, 315^\circ\}$  along with two near-identity false positives at  $15^\circ$  and  $345^\circ$ . Mean recovery over 256 models: **size 10.21, recall 89.7%, precision 71.3%**.

### 2.3 Learned-group weakness selector

For each trained model  $f$  and any candidate group  $G$ , the *invariance score* is

$$W_G(f) = \frac{1}{|G|} \sum_{g \in G} \frac{1}{|X_{\text{OOD}}|} \sum_{x \in X_{\text{OOD}}} |f(g \cdot x) - f(x)|$$

The *learned-group weakness selector* uses  $W_{G_{\text{learned}}}(f)$  where  $G_{\text{learned}}$  is inferred from training data with no oracle access. We compare against:

- $W_{G_{\text{oracle}}}(f)$  — oracle  $Z_8$  (upper bound);
- $W_{G_{\text{random}}}(f)$  — same number of randomly-chosen angles from the candidate set (soft control);
- training loss, parameter  $L_2$ , Hutchinson sharpness, training accuracy (classical baselines).

## 3. Results

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### 3.1 Group recovery

Across 256 trained models with diverse architecture/optimizer/init/augmentation:

| Statistic                    | Value        |
|------------------------------|--------------|
| Mean learned-group size      | 10.21        |
| Mean recall vs oracle Z_8    | <b>0.897</b> |
| Mean precision vs oracle Z_8 | <b>0.713</b> |

The procedure is deterministic given training data, so all 256 trials use the same per-split inference. The numbers above average across splits.

### 3.2 OOD generalization correlations

| Predictor                            | Pearson r     | Spearman $\rho$ |
|--------------------------------------|---------------|-----------------|
| <b>weakness_oracle</b> (with oracle) | <b>+0.736</b> | <b>+0.677</b>   |
| <b>weakness_learned</b> (no oracle)  | <b>+0.662</b> | <b>+0.604</b>   |
| weakness_random (soft control)       | +0.551        | +0.553          |
| parameter_l2                         | +0.429        | +0.407          |
| train_accuracy                       | +0.420        | +0.281          |
| final_train_loss                     | -0.395        | -0.291          |
| sharpness_proxy (Hutchinson)         | +0.220        | +0.283          |

**The headline:** learned-group weakness retains **90% of the oracle's predictive signal** without oracle access. Both dominate every classical predictor by 1.5× or more.

### 3.3 Per-augmentation breakdown

| Augmentation     | n  | Mean OOD | Mean $w_{\text{learned}}$ | Mean $w_{\text{oracle}}$ |
|------------------|----|----------|---------------------------|--------------------------|
| full_rotation    | 56 | 0.834    | 0.846                     | 0.903                    |
| partial_rotation | 65 | 0.705    | 0.742                     | 0.758                    |
| wrong_permute    | 68 | 0.281    | 0.478                     | 0.438                    |
| none             | 67 | 0.270    | 0.484                     | 0.403                    |

The monotone relationship is preserved: augmentations that approximately respect the symmetry produce models with high OOD and high weakness under *both* the oracle and the learned group.

### 3.4 Honest negative: the random-rotation control is not a strict null

`weakness_random` has Pearson +0.551. This is **not the clean null we wanted to report**. The reason is mechanical: with 24 evenly-spaced candidate angles, a random subset of size 10 covers 41% of the candidate space; under our 7.5° matching tolerance, ~33% of random angles fall within tolerance of a true  $Z_8$  angle by chance. So the "random-rotation control" inherits some of  $Z_8$ 's structure passively.

The cleaner null was already in [2]: `weakness_wrong_group_norm` under random *pixel permutations* (not random rotations) had Pearson −0.341, correctly anti-correlated with OOD. The interpretation we offer: rotation-from-a-dense-grid is a *soft* control that bounds learned-group weakness from below; pixel-shuffle is a *hard* control that establishes the floor. Both confirm that the cyclic-rotation structure is what is load-bearing — but only the second one is a strict null.

## 4. Limitations and Negative Results

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1. The candidate transformation set is hand-specified (we tell the procedure "consider 24 evenly-spaced rotations"). A fully end-to-end version would discover both the parameterization and the parameter range from the data.
2. The procedure is enumerative. A neural version like van der Ouderaa et al. [6] is needed for non-enumerable groups: continuous  $SO(3)$ , paraphrase substitutions on a large vocabulary, abstract algebraic invariants.
3. Pixel-space cosine similarity is a crude similarity function. On natural images with texture variation, a learned feature space is likely needed.
4. We tested only the partially-observed  $Z_n$  case. Dihedral,  $S_n$ , and product groups remain open.
5. The behavioral evidence is correlational. We have not causally intervened — e.g., retrained models *with* the learned-group data augmentation and verified OOD improves correspondingly. This is the natural next experiment.
6. The random-rotation control is partial, as discussed in §3.4.

## 5. Reproducibility

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```
# Local 64-model sweep
python3 -m experiments.learned_symmetry.sweep \
  --n-models 64 --epochs 250 --candidates 24 --threshold 0.5 \
  --base-seed 20260609 \
  --out artifacts/learned_symmetry/sweep_v1.json

# Modal-parallel 256-model sweep (used in this paper)
doppler --scope /Users/jawaun/superoptimizers run -- \
  uvx --python 3.12 --from modal modal run \
  experiments/learned_symmetry/modal_sweep.py \
  --n-shards 8 --models-per-shard 32 --epochs 250 \
  --base-seed 20260609 \
  --out artifacts/learned_symmetry/modal_sweep_v1.json

# Unit tests
python3 -m unittest discover -s tests -p "test_learned_symmetry*"
```

## 6. Discussion

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The prior paper [2] established that weakness predicts OOD generalization given oracle symmetry access. Reviewers correctly objected: *what if you don't have the oracle?* This work addresses that objection for the finite-group enumerable case. The data alone, in the partial-orbit regime that Perin and Deny [4] prove

defeats conventional supervised networks, contains enough signal to recover the symmetry group at the granularity needed for the weakness selector. Group recall 89.7% and precision 71.3% give a selector that retains 90% of the oracle's Pearson correlation with OOD.

The honest negative on the random-rotation control matters: it warns that "control" is not a category, only a relation to a specific structure. A random subset of a dense candidate set is *not* unrelated to the true group; it is partially aligned with it by construction. The pixel-permutation control from [2] is unrelated to rotation by construction and is therefore a strict null. Future versions of the benchmark should default to multiple controls rather than one.

The natural follow-on experiments are: (i) replace pixel-space cosine with a learned feature space and re-run on rotated MNIST/CIFAR; (ii) extend to dihedral and product groups with appropriate candidate sets; (iii) replace enumeration with a neural generator [6] for non-enumerable symmetries; (iv) causally validate by retraining with the learned group as data augmentation; (v) test whether the same procedure, with paraphrase substitutions as the candidate set, recovers the meaning-preserving substitution group on a small language model.

## 7. References

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- [2] **Brown, J.** *Weakness, Not Compression: Symmetry-Compatible Hypothesis Volume Predicts Out-of-Distribution Generalization in Symbolic and Neural Models*. Companion paper (2026). Establishes the weakness  $\leftrightarrow$  OOD correlation with oracle symmetry access; introduces the rotated-stroke partial-orbit benchmark.
- [3] **Cohen, T. and Welling, M.** Group Equivariant Convolutional Networks. *ICML* (2016).
- [4] **Perin, A. and Deny, S.** A Neural Kernel Theory of Symmetry Learning. arXiv:2412.11521 (2024). Proves conventional networks cannot extrapolate partially-observed cyclic symmetries; motivates our partial-orbit setup.
- [5] **Kondor, R. and Trivedi, S.** On the Generalization of Equivariance and Convolution in Neural Networks to the Action of Compact Groups. *ICML* (2018).
- [6] **Van der Ouderaa, T. F. A., van der Wilk, M., and Welling, M.** Learning Layer-wise Equivariances Automatically using Gradients. *ICLR* (2024). A neural alternative to enumerative group discovery; natural successor for non-enumerable groups.